

SE4041 – Mobile Application Design and Development

Assignment 01 – Documentation & Viva Pack

Community Services and Local Marketplace Mobile Application



Brand wordmark: *Yaluway* | Help nearby. Trade nearby. Trust nearby.



Faculty of Computing

BSc (Hons) in Information Technology | Year 4 – Semester 1 & 2

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Technology: Kotlin | XML layouts | Android Studio | Room (SQLite)

GitHub: <https://github.com/jihansandaruwan/yaluway>

Figma: <https://www.figma.com/design/0UVq1JYisRj6rl7UoadYFi/Untitled>

SDLC covered: Planning → Requirements → Design → Development → Testing

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This report follows the official Documentation marking scheme (Overview, Features, Challenges, Screenshots, Testing) and keeps the same visual style as the Phase 01 proposal and UI prototype submissions.

1. Overview of the Application

යාලේ way (Yaluway) is a Kotlin Android application built for the assigned domain Community Services and Local Marketplace (registration last digit 8). The name combines යාලේ (friend) and way (path). The tagline is Help nearby. Trade nearby. Trust nearby.

The application is a hyperlocal neighbourhood notice-board. People in the same Sri Lankan city cluster can ask for help, offer a local skill, or share an item (lend, borrow, donate, or free). It is not a payment app, chat app, or island-wide classified site. Discovery is area-first.

The implemented Android app follows the approved Figma prototype: Splash, Login, Register, three onboarding slides, Home with bottom navigation, category boards, post detail, create/edit post, saved items, incoming/sent requests, and Profile.

1.1 Problem addressed

Community help and local trade in Sri Lanka still happen on WhatsApp groups and Facebook Marketplace. Those channels are noisy, hard to search, weak on trust, and not designed for both mutual aid and neighbour trading in one place. යාලේ way puts Help, Services, and Marketplace into one structured mobile path.

1.2 Target audience

- University students who need or share everyday items (textbooks, kettle, tutor).
- Residents who need urgent or practical help (blood request, plumber, jumper cables).
- Home-based providers who offer a nearby skill (tutor, cleaner, cook).

1.3 Scope of this build

In scope (implemented)	Out of scope (not in this MVP)
Register, login, session restore	Cloud / internet backend
Create, edit, delete own posts	Online payments
Save post and send request	In-app chat
Accept / decline request + share phone	GPS live map

Nearby-area post filter	Admin dashboard
Room SQLite + form validation	Push notifications
Unit tests for validators and queries	Play Store release

1.4 Technology stack

Layer	Choice	Why
Language	Kotlin	Required by SE4041
UI	XML layouts + Material buttons	Same pattern as lab Activities / Fragments
Navigation	Activities, Fragments, bottom nav, FAB	Clear student architecture
Database	Room 2.6.1 on SQLite (yaluway.db)	Local persistence, no server
Images	Gallery picker copied to app files	Each user/post can have own photo
Onboarding	ViewPager2 with slide animation	Swipe between intro pages
Tests	JUnit unit tests	Validators and nearby-area query

2. Features and Functionalities

Features are justified against the assigned domain and the later Kotlin marking points: layouts, views, interactivity, core functionality, and database.

2.1 Feature list

Feature	What the user can do	Why it is needed
Create account	Name, email, password, Sri Lankan city	Each neighbour has their own saved account
Form validation	Red errors; Create stays off until valid	Stops numbers-in-name and fake areas
Login / session	Reopen app still logged in	Account is stored, not forgotten on close
Onboarding slideshow	Swipe Help → Services → Marketplace	Teaches the 3-in-1 model
Nearby Home feed	Sees posts near own city cluster	Negombo user does not see Kandy posts
Help / Services / Market	Open category boards	Matches the assigned 3 modules
Create post	Photo, type, mode, title, city, 10-digit phone	Publish a real neighbourhood listing
Lend / Borrow / Donate / Free	Shown only for Marketplace, with help text	Neighbour sharing, not only selling
Send request	Neighbour asks the owner	Structured contact instead of random chat
Accept / Decline	Owner shares phone only after accept	Trust: number is not public at first
Save post	Keep a listing for later	User can return to useful posts
My posts / Profile	Edit, delete, change photo and area	Account ownership and CRUD

2.2 Request procedure

This is the main social flow of the app and should be demonstrated in the viva.

- A neighbour opens a post that is not theirs and taps Send request.
- The request is stored as PENDING in the Room requests table.

- The owner sees it under Me → Requests, or on their post as View requests.
- Accept: the requester then sees the owner's phone number and can call to arrange pickup.
- Decline: the requester sees Declined. No phone number is shown.

Because there is no server, two accounts on the same phone are used to show this: log out, register a second person, send a request, then log back in as the owner.

2.3 Nearby area logic

Posts are filtered by the city saved on the logged-in account. Exact GPS is out of scope. Cities are grouped into local clusters so a Negombo user also sees Ja-Ela / Katunayake, while a Nugegoda user sees the Colombo cluster, and a Kandy user sees Kandy posts only.

The user can change city later from Me → Area settings. The Home title then reads "Posts near [city]".

2.4 Marketplace modes

- Lend — I own the item. You borrow it and must return it.
- Borrow — I need to use someone else's item for a short time.
- Donate — I give the item away. They keep it.
- Free — Anyone can take it. No money and no return.

These chips appear only when the post type is Marketplace. Help and Services hide them.

3. Development Implementation

This section maps the Kotlin build to the development marking scheme so the viva panel can see layouts, views, interactivity, functionality, and database.

3.1 Layout selection

Screen	Layout	Reason
Splash / Login / Register	ScrollView + LinearLayout	Simple forms, keyboard safe
Onboarding	ViewPager2 + page XML	Swipe slideshow
Home	NestedScrollView + RecyclerView	Banner, chips, then post list
Post lists	RecyclerView + item_post.xml	Reuse one card for Home / My posts / Saved
Create post / Detail	ScrollView + LinearLayout	Long form and action buttons
Main shell	Fragment container + bottom nav + FAB	Home / My posts / Me

3.2 UI components

- MaterialButton for primary actions (Log in, Create account, Publish, Send request).
- EditText and AutoCompleteTextView for typed fields and Sri Lankan city picker.
- RadioButton chips for Help / Services / Marketplace and Lend / Borrow / Donate / Free.
- ImageView for brand, post photos, and profile avatar.
- BottomNavigationView and yellow FAB for the main app chrome.
- RecyclerView adapters for posts and incoming/sent requests.

3.3 User interactivity

- Live red validation: Create account and Publish stay disabled until the form is correct.
- Onboarding page transformer (fade + scale) when moving from one slide to the next.
- Search box on Home filters the nearby list as the user types.

- Save toggles; Send request becomes waiting / accepted / declined.
- Tap profile photo or Choose a photo to pick a gallery image.
- Accept / Decline on the owner's request inbox.

3.4 Database integration

Room stores everything on the device in yaluway.db. There is no remote API. A SharedPreferences backup also keeps user accounts so a schema reset does not silently wipe logins.

Table	Purpose	Key fields
users	Accounts	email (PK), username, area, password, avatarPath
posts	Listings	title, category, area, contact, mode, imagePath, ownerEmail
saved_posts	Bookmarks	email + postId
requests	Ask / accept flow	requesterEmail, postId, status, createdAt

Session email is stored in SharedPreferences. Splash restores the user from Room and skips Login when a valid session exists.

3.5 Validation rules

Field	Rule
Full name	Letters, spaces and simple punctuation only. Numbers rejected.
Email	Must look like name@domain.lk / .com
Password	At least 8 characters; confirm must match
Area	Must be a city from the Sri Lankan list (type-to-search)
Post title	At least 3 characters and must contain a letter
Description	At least 10 characters
Phone	Exactly 10 digits, starting with 0

4. Challenges Faced

The assignment asks for a critical discussion of challenges and how they were solved. The points below are from the real build, not theoretical risks.

4.1 Xiaomi / Samsung USB install

Running on a physical phone failed at first with INSTALL_FAILED_USER_RESTRICTED (Installation via USB is disabled) on Xiaomi, and later the Galaxy A16 appeared in Windows as a file device but not in Android Studio until USB debugging was turned on. Solution: enable Developer options, USB debugging, and on Xiaomi also Install via USB plus a Mi account. On Samsung, File transfer + Allow USB debugging was enough.

4.2 Same picture on every profile and post

Early screens used one hardcoded drawable, so every person and every post looked the same. Solution: ImageBinder copies a gallery URI into app storage. Users and posts store imagePath. Seed posts use different drawable names so demo listings are not identical.

4.3 Accounts disappearing after close / rebuild

Room fallbackToDestructiveMigration wiped the database when the schema changed, so the same email and password looked “wrong” after a rebuild. Solution: persist users in Room and also write a SharedPreferences backup. On startup, if the users table is empty, accounts are restored. Session writes use commit() so a quick app close does not lose the logged-in email.

4.4 Unclear Marketplace options

Lend, Borrow, Donate and Free were shown for every post type, so users did not know what to pick. Solution: show those chips only for Marketplace, and put one-line help under the selected mode.

4.5 Area used as a free-text name

A full name with numbers could be typed into Area, which broke nearby filtering. Solution: autoCompleteTextView over a Sri Lankan city list. Unknown text is rejected in red.

4.6 Nearby posts without GPS

The brief is hyperlocal, but a full map API is outside the student MVP. Solution: city clusters (Negombo coast, Colombo metro, Kandy, Galle, and others). Home and category lists filter with PostQuery + SriLankaAreas.isNearby().

5. Application Screenshots

The screens below are the Figma frames that the Kotlin XML implementation follows. In the viva the same flows are shown live on the phone / emulator. Extra phone screenshots can be pasted under each figure if needed.

5.1 Entry: Splash, Login, Register

Splash shows only the brand. Login is for a returning neighbour. Register collects name, email, password, confirm password, and a selectable Sri Lankan city. The Create account button stays disabled until every field is valid. Red text appears under a wrong field.

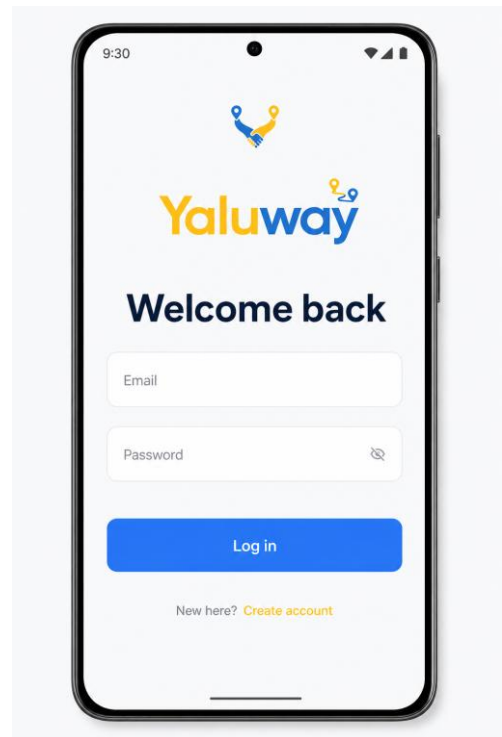
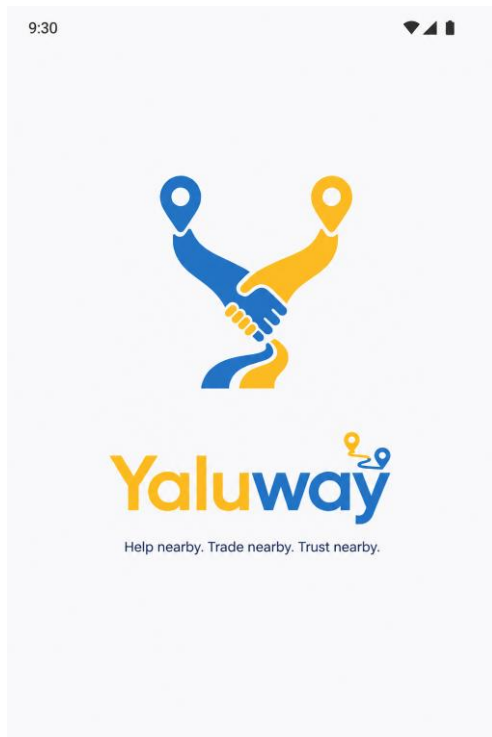


Figure 1. Splash (left) and Login (right).

 The "Create account" form is shown on a smartphone screen. At the top, there is a back arrow and the title "Create account". The form contains several input fields: "Full name" with placeholder "Enter your full name", "Email" with placeholder "Enter your email address", "Password" with placeholder "Enter your password" and an eye icon, and "Confirm password" with placeholder "Confirm your password" and an eye icon. Below these is a dropdown menu labeled "Area" with "Nugegoda" selected. A note below the dropdown says "Use your neighborhood or hostel area". A large blue "Create account" button is at the bottom of the form. The screen is framed by a black border representing a smartphone.

Figure 2. Create account — validated form with area picker.

5.2 Onboarding slideshow

After first register, three slides explain Help, Services, and Marketplace. The user can swipe or tap Next. The last slide is Get started.



Figure 3. Onboarding — Help nearby (left) and Find local services (right).

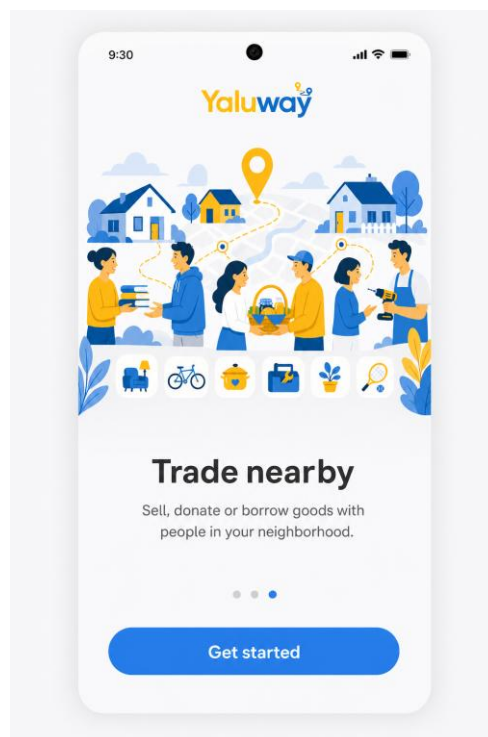


Figure 4. Onboarding — Trade nearby.

5.3 Home and category boards

Home shows the signed-in city, a search box, three module cards, and a nearby post list. Help, Services, and Marketplace open filtered boards. Cards show area, type, and time.

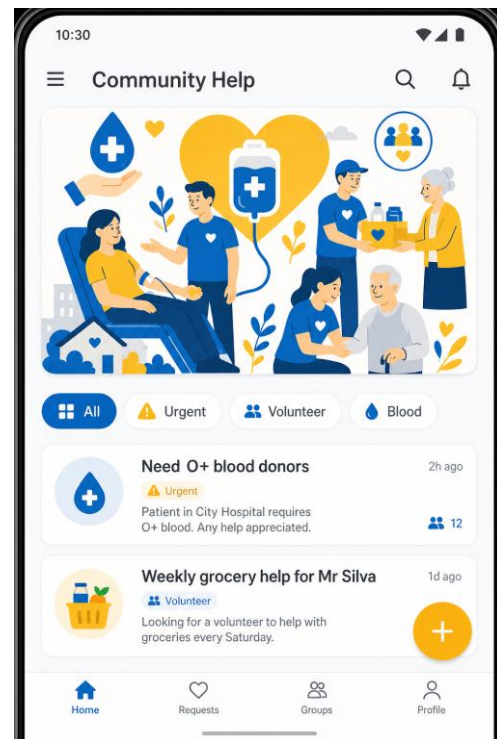
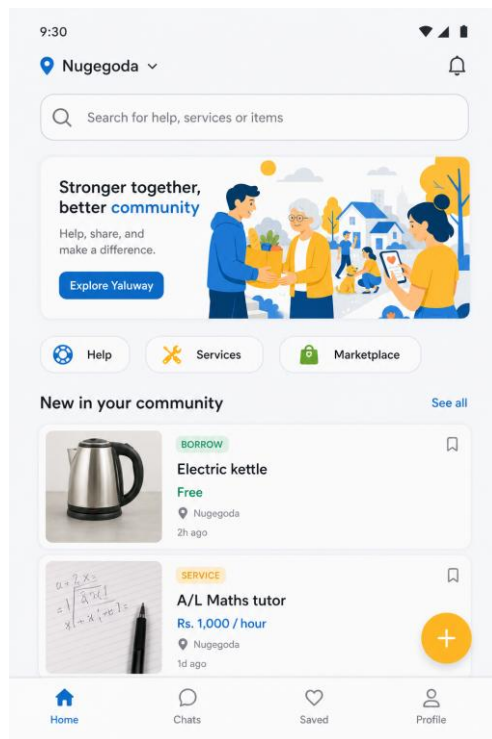


Figure 5. Home (left) and Help board (right).

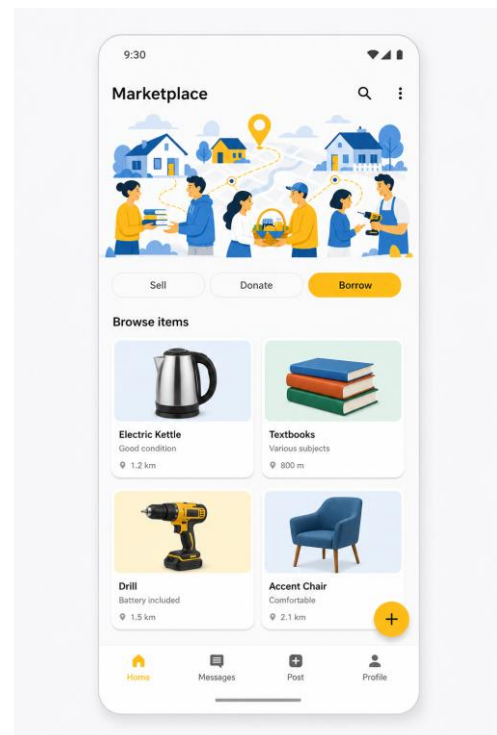
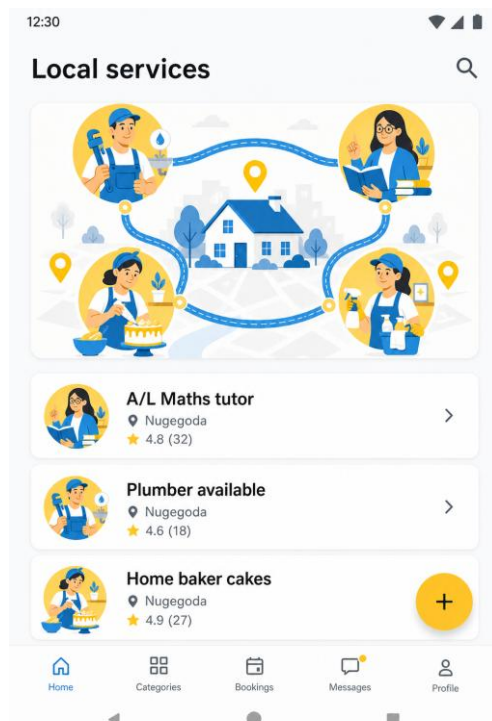


Figure 6. Services (left) and Marketplace (right).

5.4 Post detail and create post

Post detail shows the listing photo, description, and actions. A neighbour can Save or Send request. The owner sees Edit, Delete, and View requests. Create post asks for type, Marketplace mode (if needed), photo, title, description, city, and a 10-digit phone.

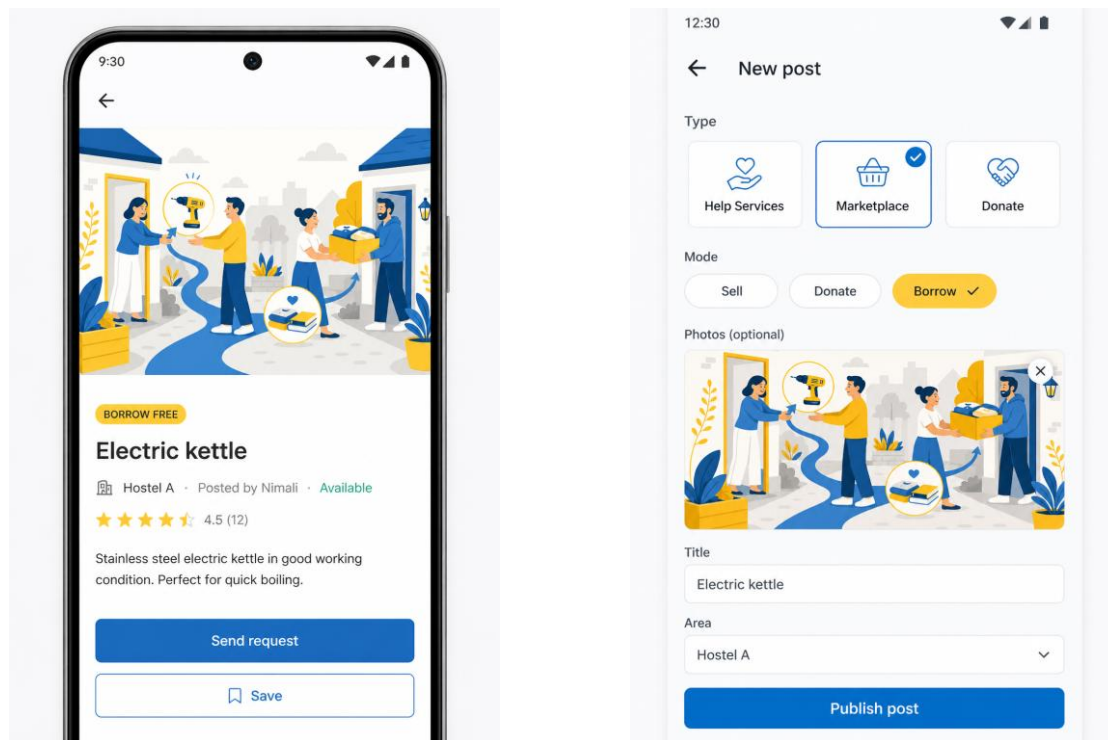


Figure 7. Post detail (left) and Create post (right).

5.5 Profile and empty state

Profile shows the user's photo (tap to change), post/saved counts, Requests inbox, saved items, area settings, and logout. Empty My posts uses a friendly illustration so a new account is not a blank screen.

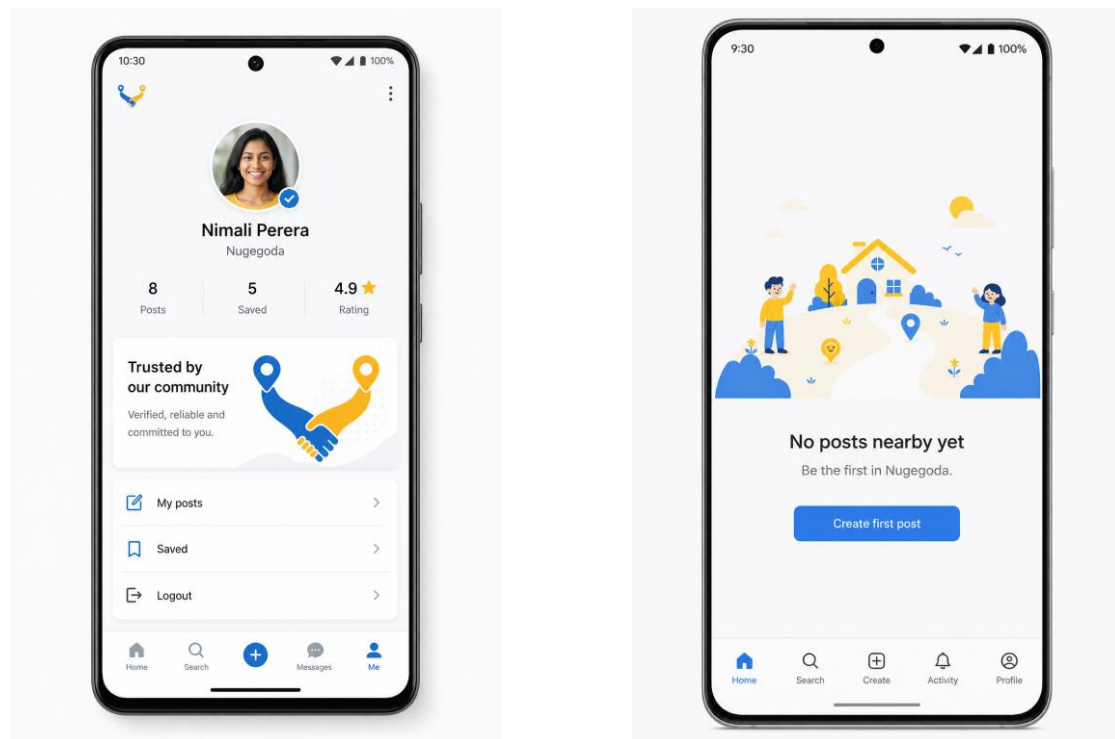


Figure 8. Profile (left) and empty My posts (right).

6. Testing Procedures and Results

Testing covers unit tests (JUnit) and integration-style checks on the running app (emulator and physical phones: Redmi / Galaxy A16).

6.1 Unit testing

Unit tests live in `app/src/test` and do not need a device. They check validation and the nearby-area query used by Home.

Test class	What it proves	Result
<code>ValidatorsTest.fullNameRejectsNumbers</code>	Kasun123 is invalid; Kasun Perera is valid	Pass
<code>ValidatorsTest.emailMustLookValid</code>	<code>kasun@yaluway.lk</code> valid; not-an-email invalid	Pass
<code>ValidatorsTest.passwordNeedsEightCharacters</code>	short rejected; 8+ accepted	Pass
<code>ValidatorsTest.phoneMustBeTenDigits</code>	0712345678 valid; 07123 invalid	Pass
<code>ValidatorsTest.areaMustBeASriLankanCity</code>	Nugegoda valid; Kasun123 not an area	Pass
<code>PostQueryTest.searchFindsArea</code>	Search “nugegoda” returns the blood post	Pass
<code>PostQueryTest.categoryFilterWorks</code>	Services filter keeps service posts	Pass
<code>PostQueryTest.urgentChipKeepsBloodPost</code>	Urgent chip keeps the blood request	Pass
<code>PostQueryTest.negomboUserSeesNearbyCoastPosts</code>	Negombo feed is coastal, not Kandy	Pass
<code>PostQueryTest.kandyUserDoesNotSeeNegomboPosts</code>	Kandy user does not see Negombo	Pass

How to run: Android Studio → Gradle → app → Tasks → verification → test, or :app:testDebugUnitTest.

6.2 Integration / manual testing

ID	Scenario	Expected	Result
IT-01	Register valid account	Session starts; onboarding then Home	Pass
IT-02	Name with a number	Red error; Create account stays off	Pass
IT-03	Area typed as a person name	Red error; must pick a city	Pass
IT-04	Close app and reopen	Still logged in, or same password works	Pass
IT-05	Log in as Negombo	Home shows Negombo / Ja-Ela posts	Pass
IT-06	Log in as Kandy	Home shows Kandy posts only	Pass
IT-07	Create Marketplace post	Photo + 10-digit phone required; modes explained	Pass
IT-08	Second account sends request	Owner sees PENDING in Requests	Pass
IT-09	Owner accepts	Requester sees phone number	Pass
IT-10	Owner declines	Requester sees Declined	Pass
IT-11	Save / unsave	Saved list updates; count on Profile updates	Pass
IT-12	Edit / delete own post	List and detail stay consistent	Pass
IT-13	Change profile photo	Home avatar and Profile update	Pass

6.3 Test evidence notes

Unit tests are code evidence. Manual tests were run on the emulator and on a Galaxy A16 after USB debugging was enabled. Xiaomi install required Install via USB. For the viva, the same scripts can be re-run live: two accounts, one request, nearby filter.

7. GitHub Repository

The brief requires the complete application via GitHub and a public repository link in this report. The project folder to upload is Assignment 01 (rootProject.name = yaluway), not the older tutorial folder.

Repository URL (replace with the live public link before printing):

<https://github.com/jihansandaruwan/yaluway>

- Keep the repository public for the viva.
- Include this documentation file and the Kotlin source.
- Do not upload app/build, .gradle, or local.properties.
- In the viva: open the GitHub page, show the package com.example.yaluway, then run the app.

7.1 Suggested commit story

The brief asks for regular commits. If the work was done locally first, still make several clear commits before the viva rather than one dump, for example: project setup, auth screens, Room database, post CRUD, requests, validation, nearby filter, documentation.

8. How to Run the Application for Viva

- Open Android Studio → Open → D:\MAD\Assignment 01.
- Wait for Gradle sync. Run on emulator or a phone with USB debugging.
- Create a first account (example: Negombo). Walk Splash → Register → Onboarding → Home.
- Show nearby posts, open a post, create a Marketplace listing with a photo and 10-digit phone.
- Log out. Create a second account in the same city. Send a request on the first user's post.
- Log back in as the owner. Open Me → Requests. Accept and show that the phone appears.
- Log in as a Kandy account to show the feed change.
- Open GitHub on the browser and point to Activities, Room entities, and unit tests.

8.1 Documents to take

- This documentation report (print or PDF).
- Phase 01 proposal: IT22294548 Sandaruwan W J - MADD proposal.pdf
- UI prototype report: IT22294548 Sandaruwan W J - MADD UI Prototype.pdf
- Public GitHub link written on the cover and ready in the browser.
- Figma prototype link ready if the panel asks about design.

9. Conclusion

ආදේශ්වය is a complete student MVP for Community Services and Local Marketplace. It implements the approved idea in Kotlin with XML screens, Room persistence, validated forms, a request workflow, and nearby-area listing. Unit tests and a manual test table document quality. The remaining viva task is to keep the GitHub repository public and demonstrate the two-account request flow plus the Negombo / Kandy nearby filter.

— End of documentation —